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Solidification in a Finned Cold Thermal Energy Storage Unit for Sustainable Cooling: Effects of Metal Foam, Hybrid Nanoparticles, and Radiative Transfer

Jasmine Nidal Abu-Hamdeh[1], Xianren Zeng[2*]

[1] Medical Internship Department, Faculty of Medicine, King Abdulaziz University, 21589 Jeddah, Saudi Arabia
[2] School of Intelligent Manufacturing, Hunan Institute of Science and Engineering, 425199 Yongzhou, China

* Correspondence: Xianren Zeng (waclin@huse.edu.cn)

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Abstract: Cold thermal energy storage supports sustainable cooling by allowing cold production and use to occur at different times, but slow solidification can restrict charging within available operating periods. This study investigates the effects of metal foam, hybrid nanoparticles, and radiative heat transfer on solidification in an enclosure with a circular cold wall and three radial fins, aiming to identify which measures most effectively shorten charging time. A transient finite element model was solved using the Galerkin method and implicit time discretization. The nanoparticle suspension was represented as a homogeneous medium, and variations in porosity, nanoparticle volume fraction, and the radiation parameter were examined through temperature distributions, solid fraction evolution, and complete solidification time. Reducing the porosity from 1 to 0.95 shortened the solidification time by approximately 81.9%. Nanoparticle addition and modeled radiative heat transfer yielded reductions of 6.54% and 8.94%, respectively, in their corresponding comparisons. Applying all three measures reduced the complete solidification time from 7742.8 to 1276.14 s, corresponding to a reduction of approximately 83.52%. Within the investigated comparisons, foam incorporation produced the largest reduction in complete solidification time, while nanoparticle addition and modeled radiative transfer produced smaller reductions. These findings identify the modifications that most strongly affect charging time in the studied enclosure and inform the selection of heat transfer measures for phase change storage in sustainable cooling applications.

Keywords: Cold thermal energy storage; Sustainable cooling; Solidification; Metal foam; Hybrid nanoparticles; Radiative heat transfer

1 Introduction

Sustainable cooling requires coordination between energy availability and cooling demand, in addition to efficient cold production. Refrigeration and air-conditioning loads vary over time, and periods suitable for producing cooling do not necessarily coincide with peak demand. Cold thermal energy storage allows cooling to be produced and stored in advance, providing flexibility in when refrigeration equipment operates. Phase change storage offers one means of achieving this separation by storing and releasing energy through solidification and melting. Recent reviews have examined storage control strategies, material selection, and the thermal constraints affecting cold storage applications [1–3]. Their findings place material properties, device-level heat transfer, and operating strategy within the same design problem. A storage unit intended for sustainable cooling must therefore be assessed by both its storage capacity and its ability to complete charging and discharging within the available operating periods.

This operational requirement makes the rate of heat transfer an important link between material performance and system application. A phase change material (PCM) may have an appropriate transition temperature and substantial latent heat yet remain unsuitable for a particular duty if charging takes too long. Experimental and numerical investigations of building cooling storage, together with assessments of shell-and-tube cold storage systems, have examined thermal response under different operating conditions [4, 5]. Such work illustrates why nominal storage capacity alone does not describe the cooling capacity accessible within a prescribed operating period. In water-based storage charged by freezing, heat must be transferred from the liquid and the advancing solidification front to the

cooling surface. The temperature field and heat transfer paths change as freezing proceeds. Shortening the complete solidification time consequently requires modifications that address these evolving thermal resistances.

Research addressing this limitation has followed two main routes: modifying the internal structure of the storage unit and modifying the storage medium. Fins and porous conductive matrices alter the distance and direction of heat transfer, whereas dispersed particles change the effective thermophysical properties of the PCM. Reviews have examined these approaches individually and in combination [6, 7], while studies of PCM thermal conductivity have further established the importance of heat transport to storage rate [8]. These routes act at different scales. Structural modifications create conductive paths through the enclosure, while material modifications alter the thermal response of the medium occupying the remaining volume. Understanding their respective roles is necessary when selecting combined measures, since adding more conductive material does not automatically produce a proportionate improvement in device performance.

Among structural approaches, fins and open-cell metal foam provide distinct ways of connecting the cooling surface to the storage medium. Fins extend the heat transfer surface into the enclosure, while metal foam forms a distributed conductive skeleton within the PCM. Buonomo et al. [9] reviewed the thermal characteristics and modeling of metal foam–PCM composites, including the influence of foam structure. At the device scale, Chen et al. [10] investigated solidification in a cold storage unit combining straight fins and metal foam. Du et al. [11] examined melting in fin–foam composite structures, and Zhang et al. [12] investigated branch-structured fins combined with nanoparticles during solidification in a triplex-tube heat exchanger. These studies demonstrate how localized fins and distributed conductive networks can modify phase change heat transfer. Their different geometries, materials, and thermal conditions also mean that reported reductions in phase change time cannot be transferred directly from one configuration to another.

The amount and placement of conductive material introduce a further design consideration because these additions occupy space otherwise available to the PCM. Configurational studies of composite PCMs have examined how material placement affects solidification in triplex-tube heat exchangers [13]. Investigations of metal foam parameters and fin–foam structures have extended the assessment to complete melting–solidification cycles [14, 15]. For ice storage, Xing et al. [16] examined different copper foam arrangements experimentally and theoretically, while Chen et al. [17] investigated partial foam filling during solidification. A subsequent numerical study compared fins, metal foam, and their combination over a complete storage cycle [18]. Porosity therefore affects more than effective conductivity: it also determines the relative volumes of the conductive skeleton and the phase change medium. A shorter solidification time must be interpreted alongside these changes before it can inform storage design.

Nanoparticle addition offers a separate means of modifying thermal response, although its effectiveness depends on dispersion stability and phase change behavior as well as conductivity. Sundaram et al. [19] examined the preparation, stability, and thermal behavior of water–graphene nanoplatelet suspensions. Their related investigation considered nucleating and thickening agents during phase change [20]. Srinivasan and Ponnusamy [21] studied the stability, thermophysical properties, and solidification characteristics of water-based hybrid suspensions containing oxygen-functionalized graphene nanoplatelets and CuO. They also investigated the influence of surfactants on CuO suspensions [22]. These experimental studies show why particle composition and formulation must be evaluated under the intended operating conditions. At the storage-device scale, numerical research on encapsulated nanoparticle-containing PCMs has considered particle concentration together with coolant operating conditions [23]. The additional improvement obtainable from nanoparticles in an enclosure that already contains fins or metal foam thus requires a configuration-specific comparison.

Combined structural and material modifications have already been investigated in numerical cold storage research. Sheikholeslami [24] examined porous foam together with hybrid nanomaterials, while Almarashi et al. [25] analyzed transient heat transfer during solidification in a nanomaterial-containing storage unit. Further studies incorporated radiative transfer into the thermal model. Osman Birkea et al. [26] considered hybrid nanoparticles, porous media, and a radiation term, and Alazwari et al. [27] investigated a finned porous enclosure containing nanomaterials. Tashkandi et al. [28] likewise included porous foam, hybrid nanoparticles, and radiative transfer in their solidification analysis. Related simulations have addressed irregular storage geometries and nanoparticle-containing water [29, 30]. These investigations establish a basis for combining the measures. The remaining task for the configuration examined here is to determine their respective contributions and relate the computed changes to practical choices in storage design.

A useful comparison must distinguish the response to each factor from the performance obtained when the measures are applied together. Percentage reductions calculated against different reference cases cannot be added directly or used alone to establish a synergistic effect. Temperature and solid fraction histories must also be interpreted alongside spatial fields and front progression to explain where and how the solidification process changes. For radiative transfer, the physical interpretation must follow the governing formulation and boundary conditions; radiation within the modeled medium is distinct from heat rejection to an external radiative sink. Against this background, three connected questions guide the present work: how foam porosity changes heat transfer and front

progression, how nanoparticle concentration and the radiation parameter affect the thermal response, and which factor produces the largest reduction in complete solidification time.

This study addresses these questions for a water-based cold thermal energy storage enclosure with a circular cold wall and three radial fins. A transient finite element model is used to examine metal foam porosity, hybrid nanoparticle volume fraction, and the radiation parameter. Temperature distributions, solid fraction evolution, solidification-front progression, and complete solidification time are compared to explain how these factors alter heat transfer and phase change rate. The contribution is a configuration-specific assessment that identifies the principal factor governing shorter solidification times and quantifies the changes associated with nanoparticle addition and radiative transfer against their corresponding reference cases. The resulting comparison links device-level heat transfer to the charging-time requirements of sustainable cooling applications and identifies which modifications warrant priority in the studied enclosure.

2 Physical Model and Numerical Method

2.1 Storage Configuration and Modeling Assumptions

The cold thermal energy storage unit consists of an enclosure with a circular inner cooling wall and three radially arranged fins, as illustrated in Figure 1. The enclosure contains a water-based phase change medium with dispersed Al_2O_3 and CuO nanoparticles. Porous metal foam is incorporated to provide additional conductive paths through the storage region. During charging, heat is transferred toward the cooling region, and solidification progresses through the surrounding medium.

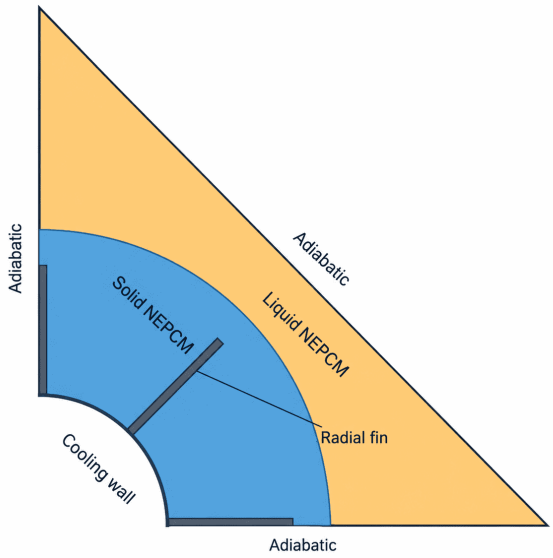


Figure 1. Schematic geometry and thermal boundaries of the finned cold storage enclosure
Note: NEPCM = nano-enhanced phase change material.

The model examines the effects of foam porosity, nanoparticle concentration, and radiative heat transfer. The porosity, γ, denotes the fraction of the foam-filled region occupied by the phase change medium. Accordingly, γ = 1 represents the absence of the metal skeleton, whereas decreasing γ increases its volume fraction. The total nanoparticle volume fraction, φ, describes the concentration of particles within the pore-filling medium. These parameters are distinguished because the metal skeleton and dispersed particles contribute differently to the effective thermal properties.

A conduction-based formulation is adopted, with buoyancy-driven motion neglected. The nanoparticle-containing medium is described using a homogeneous mixture model, so individual particle motion and spatial variations in particle concentration are not explicitly resolved. The foam-filled region is represented through effective properties rather than a pore-scale description of the metal skeleton. This treatment allows the temperature and phase state to be calculated over the enclosure without resolving individual pores or nanoparticles.

The resulting comparison concerns the thermal response of the modeled storage configuration. The homogeneous formulation does not independently describe particle sedimentation, aggregation, or changes in dispersion stability during repeated operation. Similarly, neglecting buoyancy defines the scope of the present thermal model rather than establishing that fluid motion is negligible in every physical operating condition.

2.2 Effective Thermophysical Properties

The effective properties account for the nanoparticle-containing phase change medium and the porous metal skeleton. Subscript n denotes the nanoparticle-containing medium, subscript m denotes the metal skeleton, and subscript eff denotes the homogenized foam-filled region. The conductivity and volumetric sensible heat capacity are expressed as:

$$k_{\text{eff}} = \gamma k_n + (1-\gamma) k_m \tag{1}$$

$$(\rho c_p)_{\text{eff}} = \gamma \left(\rho c_p\right)_n + (1-\gamma) \left(\rho c_p\right)_m \tag{2}$$

where, k, ρ, and c_p represent thermal conductivity, density, and specific heat capacity, respectively. Eq. (1) represents the effective thermal conductivity using a volume-weighted approximation. It describes the conductive contribution of the foam through a bulk property and does not explicitly resolve ligament geometry, tortuosity, or contact resistance. The predicted influence of porosity is therefore interpreted within this approximation.

For the nanoparticle-containing medium, subscripts w, a, and c denote water, Al_2O_3, and CuO, respectively. The particle volume fractions satisfy:

$$\phi = \phi_a + \phi_c \tag{3}$$

The density and volumetric sensible heat capacity are evaluated using mixture relations:

$$\rho_n = (1-\phi)\rho_w + \phi_a \rho_a + \phi_c \rho_c \tag{4}$$

$$\left(\rho c_p\right)_n = (1-\phi) \left(\rho c_p\right)_w + \phi_a \left(\rho c_p\right)_a + \phi_c \left(\rho c_p\right) \tag{5}$$

These expressions distinguish the total particle loading from the relative amounts of the two particle species. The thermal conductivity of the nanoparticle-containing medium, kn, enters the effective conductivity through Eq. (1).

Introducing metal foam or nanoparticles changes both the conductive properties and the proportion of phase-changing material. Consequently, a reduction in solidification time cannot be interpreted solely as an increase in thermal conductivity or as evidence of a larger latent storage capacity.

2.3 Phase Change and Governing Equations

The local solid fraction, S, describes the phase state of the storage medium. A value of zero denotes the liquid state, while unity denotes the solid state. A finite temperature interval is used to represent the transition continuously:

$$S(T) = \begin{cases} 1, & T \le T_s, \\ \frac{T_l - T}{T_l - T_s}, & T_s < T < T_l, \\ 0, & T \ge T_l \end{cases} \tag{6}$$

where, T_s and T_l are the lower and upper limits of the phase change interval. For an interval centered on the nominal transition temperature, T_m,

$$T_s = T_m - \frac{\Delta T_{\text{pc}}}{2}, T_l = T_m + \frac{\Delta T_{\text{pc}}}{2} \tag{7}$$

with $\Delta T_{\text{pc}} > 0$. This definition maintains continuity at the interval boundaries and restricts the solid fraction to physically admissible values between zero and unity.

With buoyancy-driven transport neglected, the energy balance is written as:

$$(\rho c_p)_{\text{eff}} \frac{\partial T}{\partial t} = \nabla \cdot (k_{\text{eff}} \nabla T) - \nabla \cdot \mathbf{q}_r + \mathcal{L} \frac{\partial S}{\partial t} \tag{8}$$

where, T is temperature, t is time, $\mathbf{q}_r$ is the radiative heat flux, and $\mathcal{L}$ is the volumetric latent heat coefficient, expressed in $\text{J} \cdot \text{m}^{-3}$, on the same bulk-volume basis as the other terms in the energy equation. The latent term is positive because increasing solid fraction corresponds to latent heat release.

The conductive term is retained in divergence form. For spatially uniform conductivity, it reduces to $k_{\text{eff}} \nabla^2 T$; the divergence form also makes the conservation statement explicit when properties vary between regions.

The radiative heat flux is represented by the Rosseland approximation:

$$\mathbf{q}_r = -\frac{4\sigma}{3\beta_R} \nabla T^4 \tag{9}$$

where, σ is the Stefan–Boltzmann constant and β_R is the effective Rosseland extinction coefficient. Linearization about an absolute reference temperature, T_{ref}, gives:

$$T^4 \approx 4T_{\text{ref}}^3 T - 3T_{\text{ref}}^4 \tag{10}$$

The reference temperature is distinguished from the numerical phase change interval in Eq. (7). Substitution into Eq. (9) gives an equivalent radiative conductivity:

$$\mathbf{q}_r = -k_r \nabla T, k_r = \frac{16\sigma T_{\text{ref}}^3}{3\beta_R} \tag{11}$$

The dimensionless radiation parameter is defined as:

$$Rd = \frac{16\sigma T_{\text{ref}}^3}{\beta_R k_w}, k_r = \frac{Rd}{3} k_w \tag{12}$$

The energy equation consequently becomes:

$$(\rho c_p)_{\text{eff}} \frac{\partial T}{\partial t} = \nabla \cdot [(k_{\text{eff}} + k_r) \nabla T] + \mathcal{L} \frac{\partial S}{\partial t} \tag{13}$$

In this formulation, radiation contributes to internal heat transport. It does not introduce a separate external cooling boundary. The Rosseland approximation is conditional on an optically thick participating medium, and its applicability depends on the optical properties, characteristic dimensions, and boundary treatment of the modeled system.

2.4 Numerical Solution and Evaluation of Thermal Response

The circular inner wall provides the cooling boundary, while the boundaries identified as adiabatic in Figure 1 prevent heat exchange with the surroundings. The initial temperature determines the initial phase state through Eq. (6), ensuring consistency between temperature and solid fraction.

The computational domain is discretized using triangular finite elements. The governing energy equation is solved using a Galerkin formulation, and the transient terms are treated implicitly. This approach couples the evolving temperature field to the phase state through the solid fraction relation. An adaptive mesh is used to provide additional spatial resolution near regions of rapid thermal variation and the advancing solidification zone. The conduction-based formulation follows the general modeling approach discussed in the related cold storage literature [24].

The numerical results are examined using temperature distributions, solid fraction fields, and solidification-front progression. These quantities provide complementary information: temperature distributions describe the development of thermal gradients, while solid fraction fields indicate how rapidly different parts of the enclosure undergo phase change. Their combined interpretation helps distinguish local changes near the cooling region from changes across the storage volume.

Complete solidification time is used to compare the thermal response of different configurations. For a reference time t_{ref} and a comparison time t_f, the relative reduction is:

$$R_t = \frac{t_{\text{ref}} - t_f}{t_{\text{ref}}} \times 100\% \tag{14}$$

Each percentage reduction is associated with its corresponding reference condition. Comparisons intended to isolate a particular parameter require the remaining inputs to be held constant. The combined configuration is evaluated against its stated baseline, and reductions calculated from different reference cases are not added or interpreted as independent contributions to a common total.

The assessment addresses the rate of cold charging within the modeled enclosure. Faster solidification can help reduce the time required to charge a phase change storage unit, but it does not independently establish lower refrigeration electricity consumption or reduced environmental impact. Its relevance to sustainable cooling is therefore discussed in terms of the charging-time constraint addressed by the investigated heat transfer measures.

3 Results and Discussion

3.1 Numerical Representation and Overall Solidification Behavior

Figure 2 illustrates the triangular mesh used to represent the enclosure. The adaptive discretization provides additional resolution around regions of rapid thermal variation, including the cooling surfaces and the advancing phase change zone.

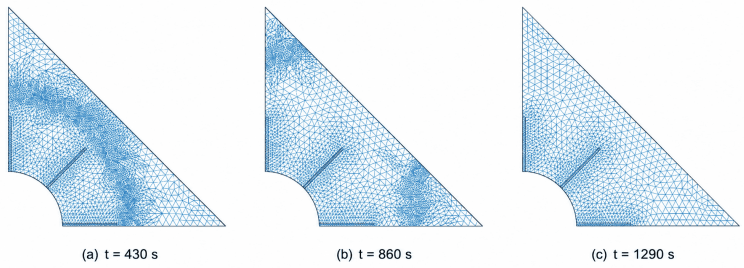


Figure 2. Computational meshes at (a) 430, (b) 860, and (c) 1290 s for $\varphi = 0.02$, $\gamma = 0.95$, and $Rd = 1$

With this numerical representation established, Figure 3, Figure 4, Figure 5, and Figure 6 show how cooling developed into solidification within the enclosure. The solid phase appeared first near the inner circular wall and the attached fins, where heat had the shortest paths to the cooling region. As charging continued, the interface moved into the surrounding medium, leaving the liquid progressively farther from the cooled surfaces. The changing interface shape therefore reflected the distribution of heat transfer paths rather than uniform freezing throughout the enclosure.

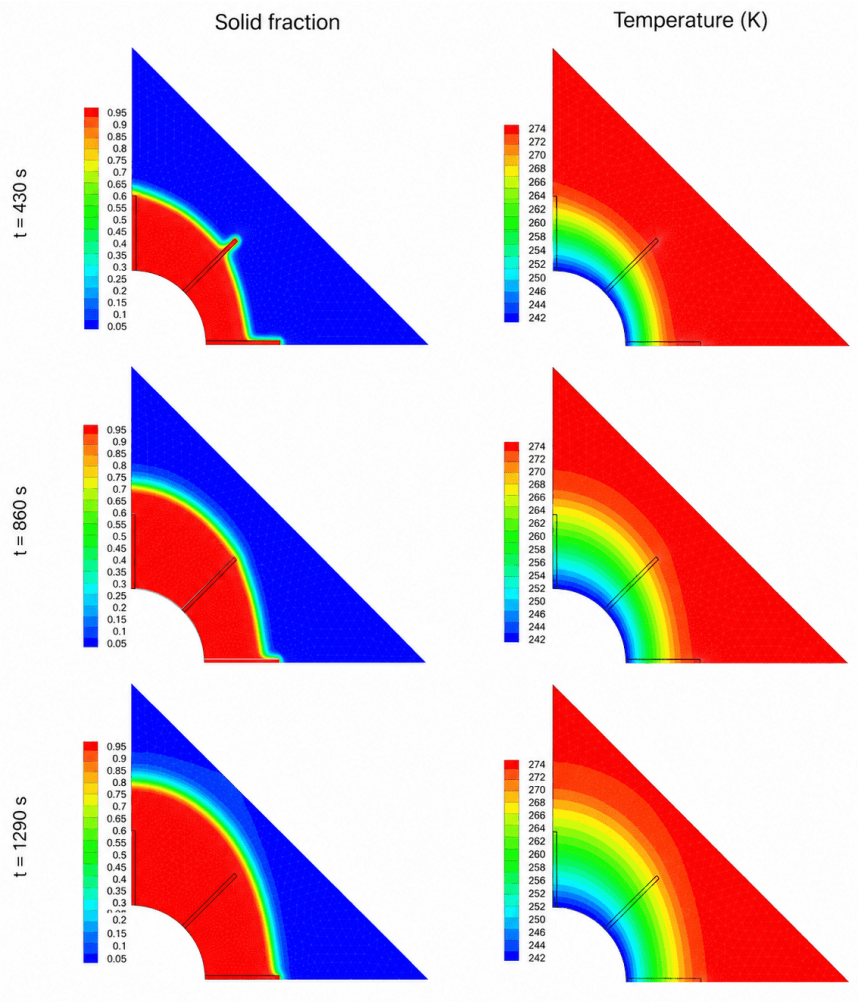


Figure 3. Solid fraction and temperature distributions at 430, 860, and 1290 s for $\varphi = 0$, $Rd = 1$, and $\gamma = 1$

The radial fins influenced this progression by extending the cooling region into the storage medium. Regions adjacent to the fins solidified earlier, whereas liquid remained longer in areas between them and near the outer adiabatic boundary. These differences explain why a description based only on mean temperature would be incomplete: similar average temperatures can coexist with different distributions of residual liquid. The temperature and solid fraction fields must therefore be considered together when assessing the approach to complete solidification.

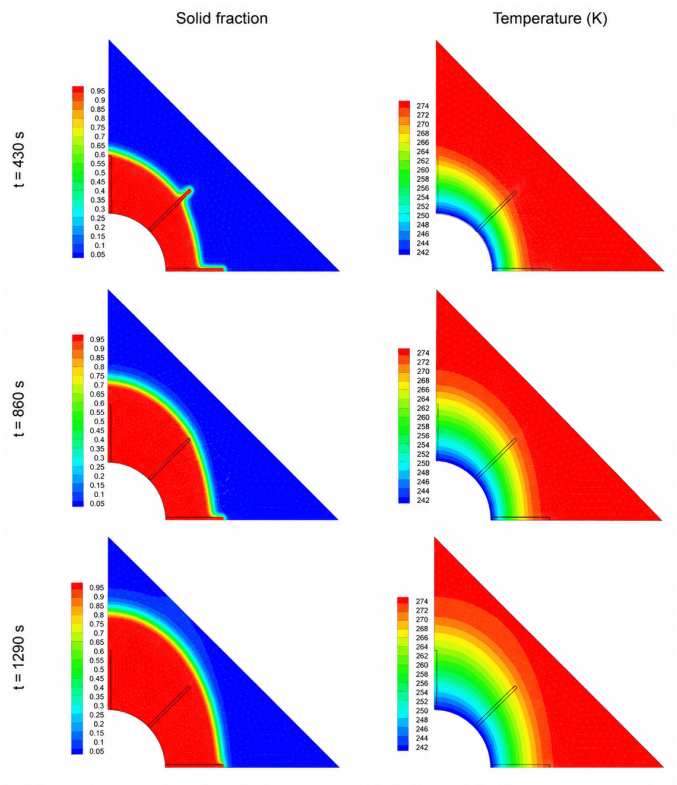


Figure 4. Solid fraction and temperature distributions at 430, 860, and 1290 s for $\varphi = 0.02$, $Rd = 1$, and $\gamma = 1$

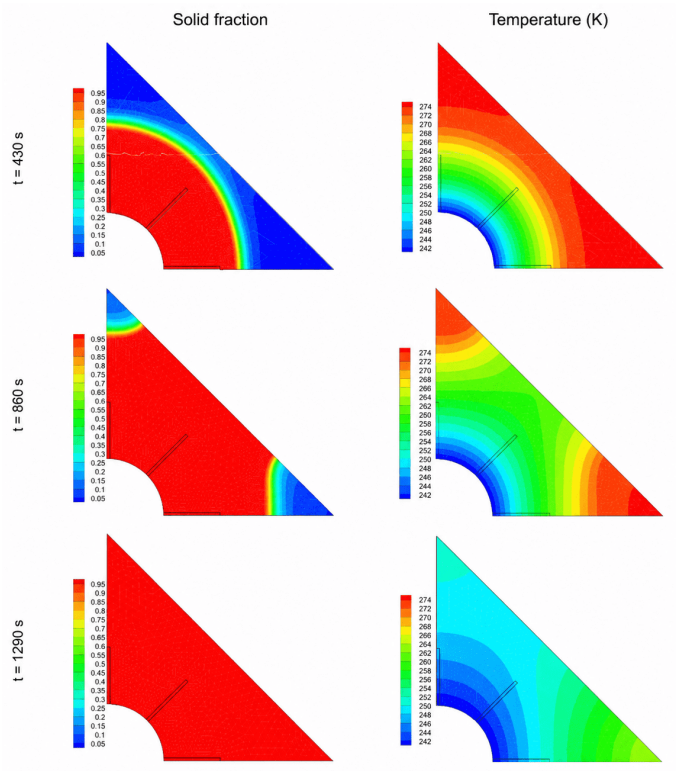


Figure 5. Solid fraction and temperature distributions at 430, 860, and 1290 s for $\varphi = 0.02$, $Rd = 1$, and $\gamma = 0.95$

Figure 7 makes these spatial differences more apparent by comparing successive front positions. At corresponding times, greater advancement into the enclosure indicated faster solidification. The porous structure and the modeled radiative contribution shifted the interface farther from the cooling region, consistently with more rapid heat removal.

The comparison therefore indicates differences in front advancement at the sampled times.

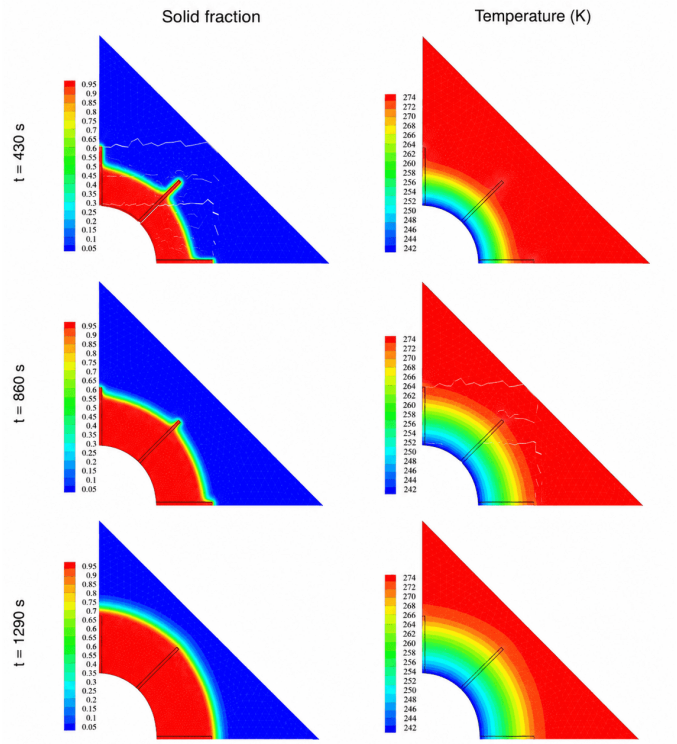


Figure 6. Solid fraction and temperature distributions during freezing for $\varphi = 0.02$, $\gamma = 1$ and $Rd = 0$

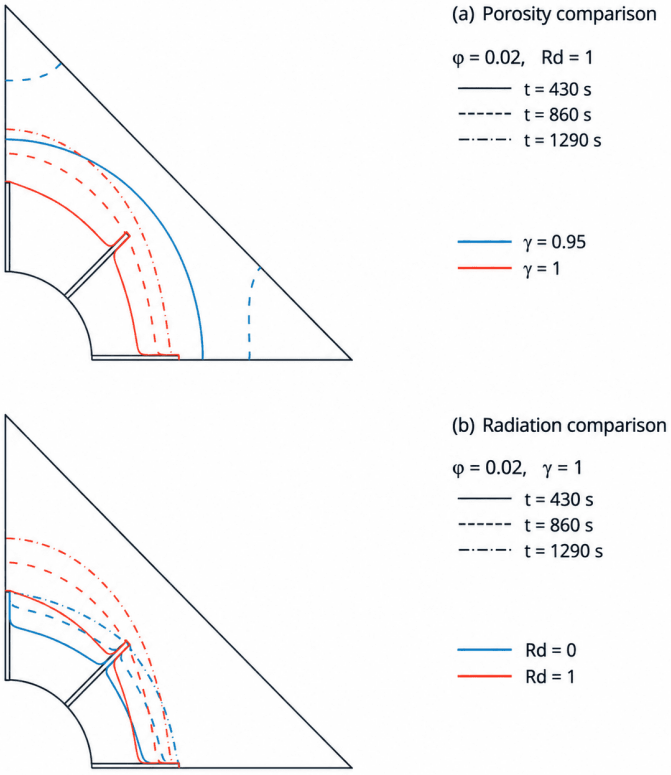


Figure 7. Solidification-front positions at different times: (a) effect of porosity; (b) effect of the radiation parameter

The common feature across these observations is the dependence of freezing on the thermal connection between

the remaining liquid and the cooling region. This provides the physical basis for the parameter comparisons that follow. Foam porosity changes the distributed conductive network, nanoparticle concentration changes the properties of the pore-filling medium, and the radiation parameter changes an additional internal transport contribution. Their effects can consequently be interpreted in terms of how each modifies the same underlying heat transfer process.

3.2 Role of the Porous Conductive Skeleton

Among these modifications, metal foam most directly changes the conductive connection across the storage volume. Figure 8 compares porosities of 0.95 and 1, with the latter representing the absence of the metal skeleton. The lower-porosity case showed a faster increase in solid fraction and a more rapid decrease in temperature. Its energy history also declined more quickly, linking the spatial advancement observed previously to a faster overall charging response.

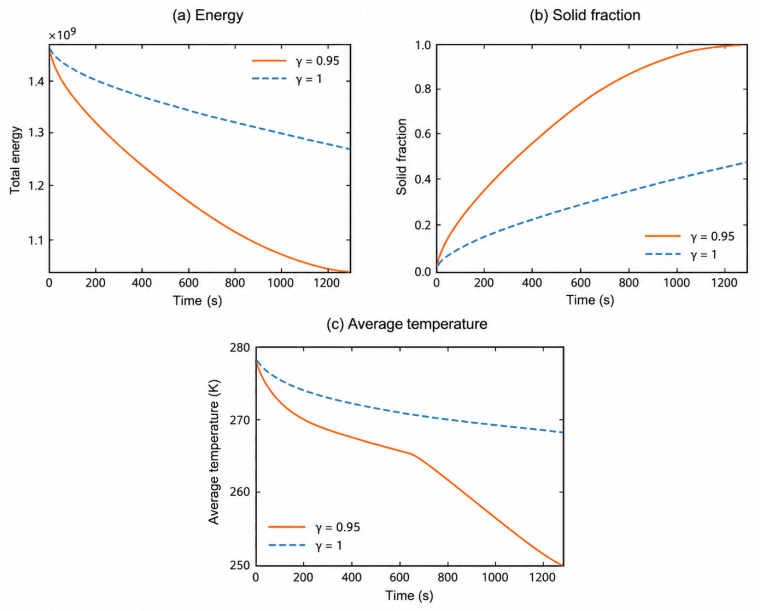


Figure 8. Time histories of (a) energy, (b) solid fraction, and (c) average temperature for different porosities

This behavior follows from the role assigned to the skeleton in the effective-property model. Decreasing porosity increases the metal volume fraction and raises the bulk conductivity when the skeleton is more conductive than the pore-filling medium. Heat can then travel through the distributed metal network as well as through the phase change medium. The foam thus extends the conductive influence of the cooling region beyond the localized paths provided by the fins.

The associated temperature and solid fraction histories describe different aspects of this response. The greater solid fraction indicates that a larger proportion of the phase-changing medium had frozen at a given time, while the lower temperature also includes sensible cooling of material that had already solidified. The energy decrease is consistent with both processes. It should therefore be interpreted as a reduction in the thermal energy remaining within the unit, rather than a decrease in the intrinsic latent heat of water.

The stronger thermal response does, however, arise in a composite containing less water. Introducing the skeleton reduces the space available to the phase change medium, so the shorter freezing period reflects the combined consequences of altered heat transport and material inventory. The comparison establishes that the modeled foam-containing configuration completed charging more rapidly; it does not establish a larger latent storage capacity.

Beyond the change in conductive structure, the thermal response also depends on the composition of the phase change medium. The following comparison therefore examines nanoparticle loading while distinguishing its effect from that of foam incorporation.

3.3 Contribution of Hybrid Nanoparticles

Figure 9 compares the particle-free medium with a total nanoparticle volume fraction of 0.02. The nanoparticle-containing case exhibited a slightly greater solid fraction and lower temperature at corresponding times, accompanied by a somewhat faster decrease in energy. The curves remained closer together than those in the porosity comparison, indicating a smaller change in thermal response for the particle-loading variation shown.

The reported complete-solidification times provide a quantitative measure of this difference. For the corresponding nanoparticle comparison, the time decreased from 7050.78 to 6589.34 s, equivalent to a reduction of approximately 6.54%. Thus, the particle addition changed the charging response, but the magnitude of that change was modest relative to the reported foam effect.

The contrast between the two comparisons is consistent with their different representations in the model. The foam introduces a distributed conductive skeleton, whereas the nanoparticles modify the effective properties of the medium occupying the pores. Particle addition does not create the same continuous, enclosure-scale path between the remaining liquid and the cooling region. The reported results therefore favor examining the conductive architecture before expecting a comparable improvement from a small change in particle concentration.

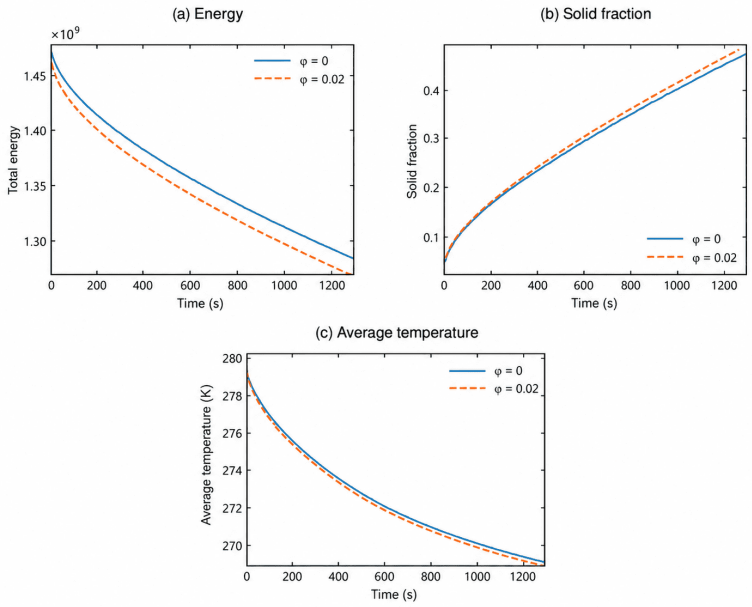


Figure 9. Time histories of (a) energy, (b) solid fraction, and (c) average temperature for different nanoparticle volume fractions

The nanoparticle response is nevertheless specific to the adopted mixture properties and loading. The homogeneous formulation does not resolve changes in dispersion stability or particle redistribution during freezing, and the particles also displace part of the water. The 6.54% reduction consequently describes the stated thermal comparison rather than an optimum concentration or a demonstrated increase in storage capacity.

Radiative transport provides a further modeled contribution to heat transfer, whose effect is examined through the temperature and solid fraction histories.

3.4 Influence of Modeled Radiative Transfer

Figure 10 compares $Rd = 0$ and $Rd = 1$. Increasing the radiation parameter produced a lower temperature, a greater solid fraction, and a faster decline in the energy history over the plotted interval. These changes were consistent with more rapid transfer of heat toward the cooling region and with the greater front advancement shown in Figure 7.

Within the adopted formulation, increasing Rd increases the radiative contribution to effective internal transport. The resulting acceleration is therefore attributed to modeled radiative heat transfer. It is not evidence of a separate external radiative cooling process, because heat rejection from the enclosure remains governed by the cooling boundary.

The reported times decreased from 7742.80 to 7050.78 s for the corresponding radiation comparison, giving a reduction of approximately 8.94%. This was larger than the 6.54% reduction in the stated nanoparticle comparison. The comparison supports a difference between these particular parameter changes, but it does not establish a universal effectiveness ratio: the percentage reduction depends on the reference state and the range over which each parameter is varied.

The radiation result also depends on assumptions that differ from those governing the conductive skeleton. The Rosseland approximation requires an appropriate participating-medium description, optical thickness, and boundary treatment. Figure 10 shows the response of the numerical model to Rd; it does not independently establish that these

optical conditions are satisfied by a physical foam–nanoparticle composite. The quantitative interpretation should therefore remain tied to the adopted radiation model.

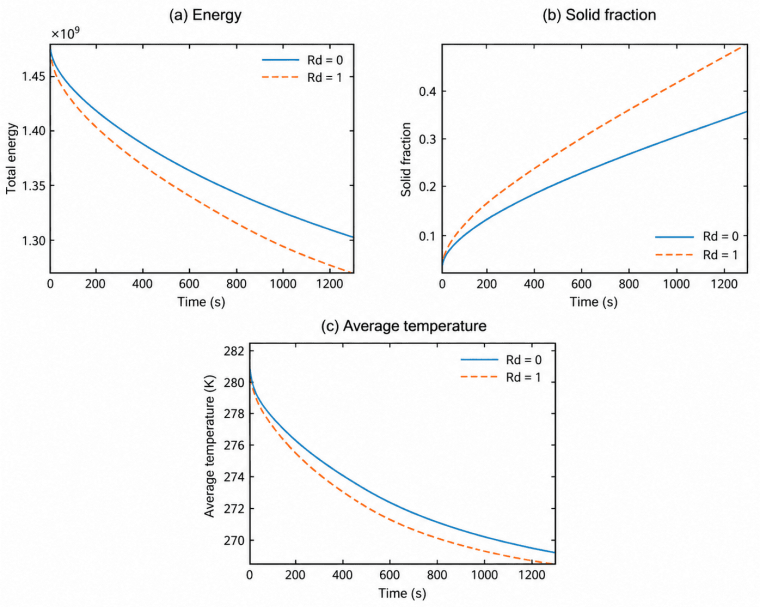


Figure 10. Time histories of (a) energy, (b) solid fraction, and (c) average temperature for different radiation parameters

The separate comparisons show how particle loading and radiative transport affect the thermal response. Figure 11 extends this assessment by presenting complete solidification time across the investigated parameter combinations, allowing these changes to be compared with the porosity-dependent response.

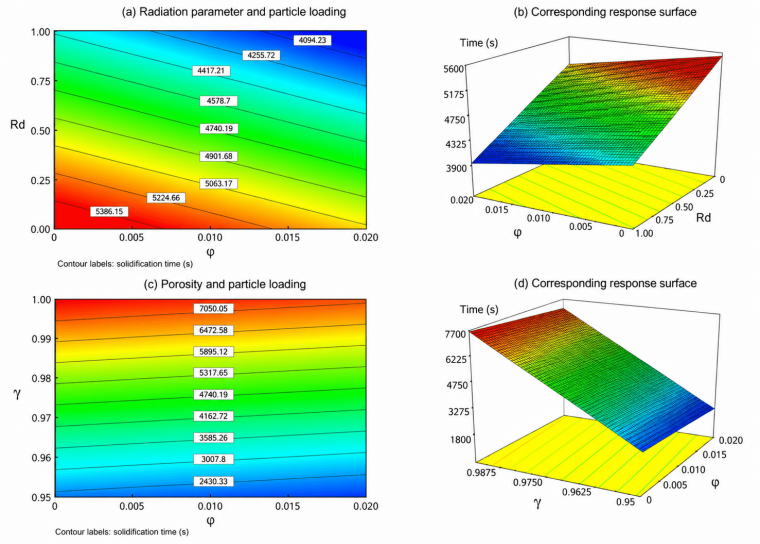


Figure 11. Complete solidification time versus nanoparticle loading and radiation parameter and nanoparticle loading and porosity: (a) contour plot for the interaction between φ and Rd; (b) the corresponding 3D response surface of (a); (c) contour plot for the interaction between φ and γ; (d) the corresponding 3D response surface of (c)

3.5 Combined Response and Implications for Storage Design

Figure 11 summarizes the variation in complete solidification time across the investigated parameter combinations. The porosity-dependent results displayed a substantially larger change than the nanoparticle-concentration variation, while increasing the radiation parameter also shortened the modeled freezing period. This ordering is consistent with the temperature and solid fraction histories in Figure 8, Figure 9, and Figure 10 which show the largest change for the investigated porosity variation.

The porosity comparison yielded an approximately 81.9% reduction in complete solidification time when γ decreased from 1 to 0.95. Such a large response follows the strong change in conductive transport represented by the bulk-property model, together with the reduction in the amount of phase-changing medium. Its magnitude is therefore a result for the specified composite representation, rather than a general performance value for metal foam.

When all three measures were applied, the reported complete solidification time was 1276.14 s, compared with 7742.80 s for the stated baseline. These values give an overall reduction of approximately 83.52%. This percentage expresses the total difference between the two configurations. It cannot be obtained by adding the separately reported reductions, because those comparisons do not necessarily share the same reference state.

The combined reduction describes the difference between the selected configurations; it does not isolate interaction effects among the three measures.

The practical outcome is a priority for design assessment. Within the modeled conditions, the distributed conductive structure produced the largest change in charging response, while nanoparticle addition and radiation produced smaller changes in their respective comparisons. This suggests that the arrangement and amount of conductive material should be evaluated before additional modifications are selected. The purpose is not simply to maximize conductivity, but to obtain an acceptable charging time while retaining the water volume needed for the required storage duty.

The shorter freezing period addresses a device-level constraint on completing cold charging within an available operating window. This is relevant to storage-assisted sustainable cooling, where cooling production and demand can be separated in time. The present comparison concerns charging duration; system electricity consumption, emissions, and lifecycle performance require separate evaluation.

4 Conclusions

This study examined solidification in a water-based cold thermal energy storage enclosure with a circular inner cooling wall and three radial fins. The effects of metal foam porosity, hybrid nanoparticle concentration, and the radiation parameter were compared to identify the principal factors controlling charging time.

Solidification began near the cooling wall and fins and progressed toward regions farther from these surfaces. Introducing metal foam accelerated this progression, with the porosity change from 1 to 0.95 reducing the complete solidification time by approximately 81.9%. This was the largest reduction among the investigated comparisons. Within the effective-property model, the response reflected both improved conductive transport through the metal skeleton and displacement of part of the phase-changing medium.

Nanoparticle addition and modeled radiative transfer produced smaller reductions of 6.54% and 8.94%, respectively, in their corresponding comparisons. With all three measures present, the complete solidification time decreased from 7742.80 to 1276.14 s, corresponding to an overall reduction of approximately 83.52%. These percentages describe their stated reference cases and are not additive measures of a common total.

The main design implication is that the distributed conductive structure warrants priority when seeking to shorten charging in the studied enclosure. Nanoparticle addition and radiative transport modify the thermal response, but their contribution must be assessed alongside the selected foam configuration and the amount of water retained for latent heat storage. This distinction connects faster solidification to a practical charging-time requirement without equating it with increased storage capacity.

The findings provide a configuration-specific basis for selecting heat transfer modifications for cold storage intended to support sustainable cooling. Their interpretation remains conditional on the conduction-based formulation, effective-property relations, and radiation assumptions. System-level electricity savings and environmental performance were not evaluated. Further assessment should establish whether the shorter charging period is retained under experimentally supported material properties and operating conditions.

Author Contributions

Both authors contributed equally to this work, with each accounting for 50% of the overall contribution. Both authors read and approved the final manuscript.

Data Availability

The data supporting the findings of this study are available from the corresponding author upon reasonable request.

Conflicts of Interest

The authors declare no conflicts of interest.

Declaration on the Use of Generative AI and AI-assisted Technologies

AI-assisted tools were used during manuscript preparation to revise the language and improve formatting. The authors reviewed and edited the resulting text and take full responsibility for the content of the published article.

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